

Vitalii Smekalov

Python Backend Developer | 3 years of commercial development | remote

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SUMMARY

I build backends in Python: models and REST APIs with Django REST Framework, background jobs with Celery, and a React frontend when the task needs one. For three years I have worked with enterprise data in Oracle and PostgreSQL.

EXPERIENCE

Lead Engineer (Python service development)

Aug 2023 - present (3 years 2 months)

MegaFon, telecommunications | remote

Enterprise data warehouse operations team. I develop internal data quality services for analysts and the on-call shift on top of Oracle DWH and Greenplum.

- **Data mart readiness forecasting service** (Django, Celery, PostgreSQL). Designed and built it from scratch: collecting scheduler history from Oracle (about 270 tasks, a dependency graph of 68k links), a quantile-based statistical model that accounts for the calendar and task dependencies, and forecast updates from the current day's facts. Mean error in production is 33 minutes versus 123 for an estimate based on the historical norm.
- Owned the project from requirements to release: gathered requirements from the on-call shift and analysts, documented the architecture decision (ADR), presented the results in a talk.
- Set up quality control for the model: backtesting on historical data with MAE and p90 coverage metrics, about 170 unit tests. Changes to the forecast logic are accepted only on backtest results.
- **Data quality portal** (Django REST Framework, React, Redux): dashboard and KPI checks, HTML email reports, SSO via Keycloak (JWT), access control by Active Directory groups (LDAP). The portal replaced manual metric reconciliation through SQL and email.
- Background jobs on Celery with RabbitMQ as the broker, schedules in Redis (redbeat), idempotent processing for redelivered tasks. Services run in Kubernetes; Docker image builds and CI/CD.
- Integration with the enterprise warehouse: Django ORM over a legacy Oracle schema, several databases in one project (PostgreSQL, Oracle, Greenplum), loading reference data from CSV with fuzzy record matching.
- Reviewed the portal codebase and drew up a prioritized refactoring plan. Investigated ETL incidents (Oracle, ODI, PL/SQL) down to the root cause.

SIDE PROJECT

open-lumen: an end-to-end encrypted messenger. Go backend, React and TypeScript frontend. github.com/kultvio

SKILLS

Languages: Python 3, SQL, PL/SQL, Go, JavaScript, TypeScript

Backend: Django, Django REST Framework, FastAPI, Celery, RabbitMQ, Redis, REST API, JWT, OAuth 2.0, Keycloak, LDAP

Databases: PostgreSQL, Oracle, Greenplum, Oracle Data Integrator (ODI)

Frontend: React, Redux

Tools: Git, Docker, CI/CD, Nginx, Linux, unittest

EDUCATION

Pacific National University, Khabarovsk, Russia. Computer Security.

LANGUAGES

Russian: native. English: B2.